



SYDNEY BOYS HIGH
SCHOOL
MOORE PARK, SURRY HILLS

2007
HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK #3

Mathematics

(2 Unit)

General Instructions

- Reading Time – 5 Minutes
- Working time – 120 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- All necessary working should be shown in every question.
- Hand in your answers in 3 separate bundles.
Section A (Question 1 and Question 2),
Section B (Question 3 and Question 4) and
Section C (Question 5 and Question 6)

Total Marks – 100

- Attempt questions 1-6
- All questions are **NOT** of equal value.

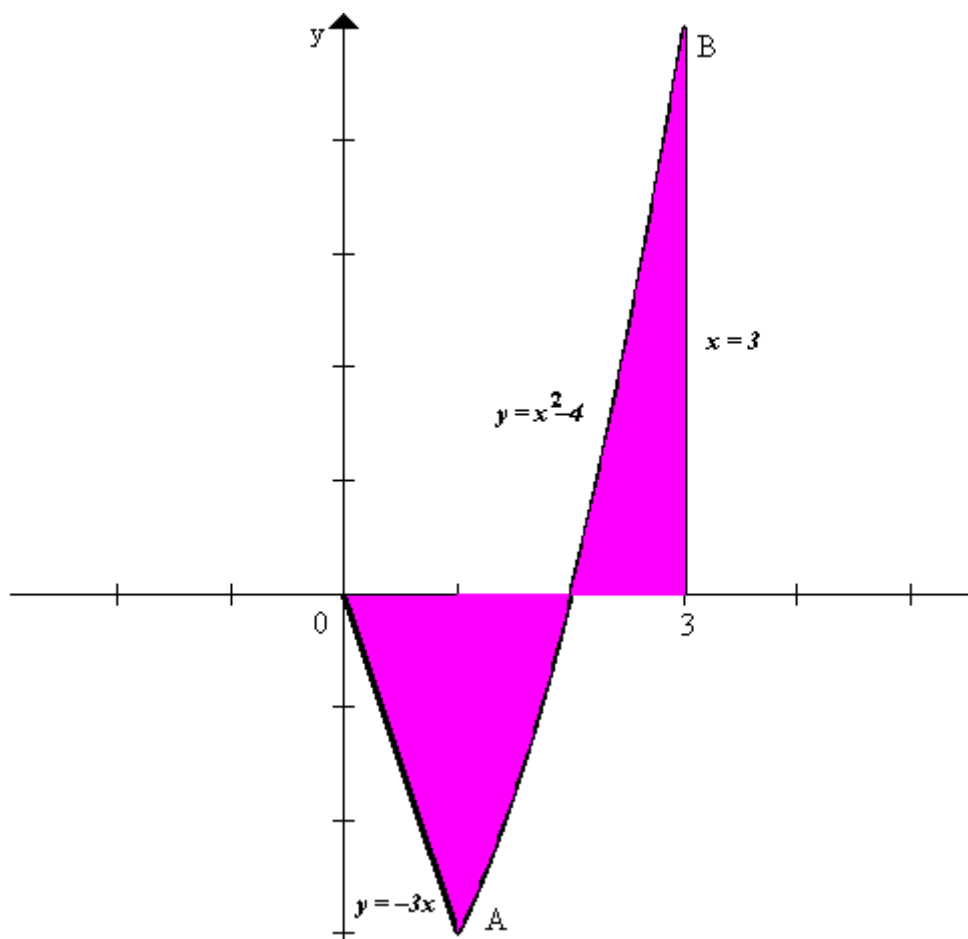
Examiner: *E. Choy*

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Section A – Start a new booklet.

Question 1 (23 marks).		Marks
a)	Differentiate with respect to x	
(i)	$\ln 3x$	2
(ii)	$\tan \frac{x}{2}$	2
(iii)	$\frac{1}{2} \sin 4x$	2
(iv)	$4 + 5e^{-x}$	2
b)	(i) What is the second derivative of $e^{0.2x}$?	2
	(ii) Find $f''(2)$ if $f(x) = 5 - 2 \ln x$	2
	(iii) Find the domain for which $y = x - 2x^3$ is concave downwards.	2
	(iv) Find the point of inflexion on the curve $y = 3x^3 - 3x^2 + 3x - 1$.	2
c)	Evaluate $\int_{-1}^0 \sin \pi x \, dx$	2

d)



The shaded region is bounded by the lines $x = 3$, $y = -3x$, the x -axis and the curve $y = x^2 - 4$.

Show that A is the point $(1, -3)$ and find the area of the shaded region of the graph.

3

e)

Sketch the graph of $y = \tan 2x$ for $0 \leq x \leq 2\pi$

2

Question 2 (15 Marks).**Marks**

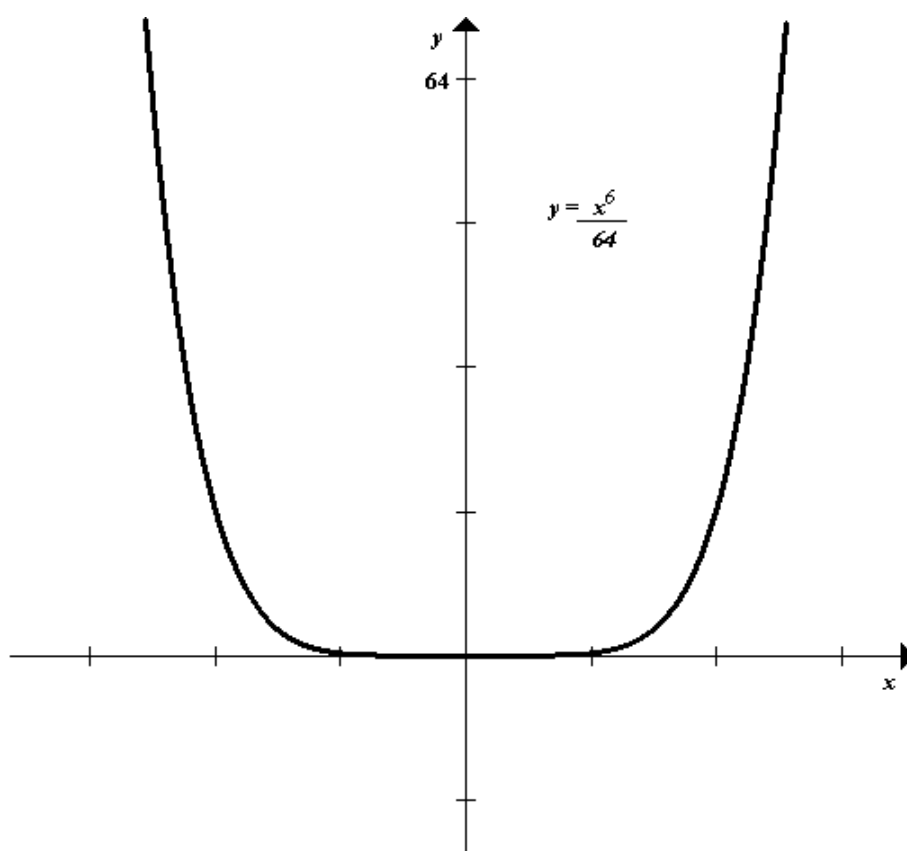
a) Solve $\log_7 64 = 3\log_7 x$

1

b) Find the exact value of $\int_{\frac{1}{3}}^{\frac{1}{2}} \sec^2\left(\frac{\pi x}{2}\right) dx$

2

c)



A bowl is formed by rotating part of the curve $y = \frac{x^6}{64}$ between $x = 0$ and $x = 4$ about the y -axis.

(i) Show that $x^2 = 4y^{\frac{1}{3}}$

1

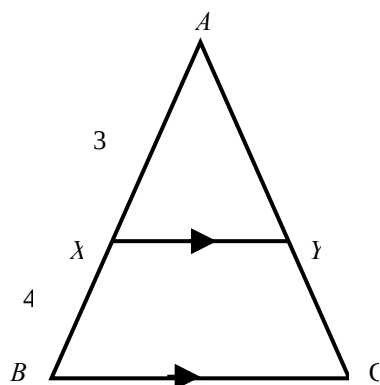
(ii) Find the volume of the bowl.

3

d) Convert 240° to radians.

1

e)



With respect to the diagram above write down the ratios:

- | | | |
|-------|-----------|---|
| (i) | $AY : YC$ | 1 |
| (ii) | $AY : AC$ | 1 |
| (iii) | $YC : AC$ | 1 |

- f) ABC is a triangle in which $AB = AC$. If P, Q and R are collinear points on AC, CB and AB produced respectively, show that $BR \times PQ = PC \times RQ$ 4

End of Section A.

Section B – Start a new booklet.

Question 3 (19 marks).

Marks

- a) (i) Copy and complete the following table of values for the curve

$$y = \frac{1}{x+1}$$

1

x	0	$\frac{1}{2}$	1	$1\frac{1}{2}$	2
y					

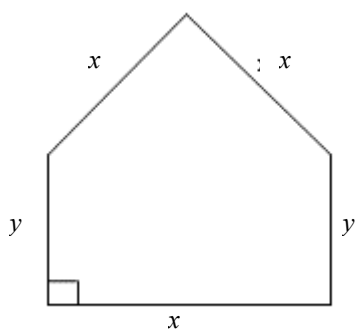
- (ii) Use the trapezoidal rule with 5 function values from part (i), to give an approximation to $\int_0^2 \frac{dx}{1+x}$. Give your answer correct to 2 decimal places.

3

- b) Given that n is a positive number, find the smallest and largest of the following numbers; $e^{\frac{n}{2}}, e^n, e^{-n}, e^{-\frac{1}{2}n}$.

1

c)



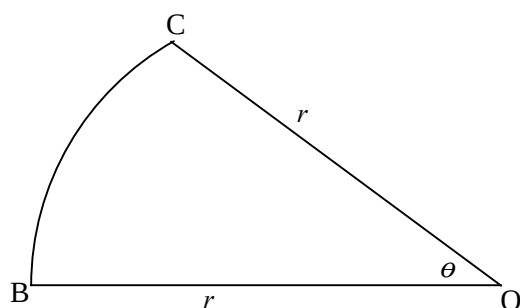
A piece of wire is bent to form a pentagon as shown. The area enclosed by the wire is 33cm^2 .

- (i) Express y in terms of x .

2

- (ii) Show that the perimeter is a minimum when $x = 2\sqrt{6+\sqrt{3}}$

4

d)

The diagram above shows a sector OBC of a circle with centre O and radius r cm. The arc BC subtends an angle θ radians at O .

(i) Show that the perimeter of the sector is $r(2 + \theta)$. 2

(ii) Given that the perimeter of the sector is 36cm, show that its area

$$A = \frac{648\theta}{(\theta + 2)^2} \quad \text{2}$$

(iii) Hence show that the maximum area of the sector is 81cm^2 . 4

Question 4 (14 marks).**Marks**

- a) Differentiate $x^2 \ln x$ and hence determine $\int_{\sqrt{e}}^e x \ln x \, dx$ 4
- b) The region R is bounded by the curve $y = \tan x$, the x -axis and the vertical line $x = \frac{\pi}{3}$
- (i) Sketch R and find its area. 2
- (ii) Find the volume generated when R is rotated about the x -axis. 2
- c) Consider the function $y = 1 + e^{2x}$
- (i) What is the domain of the function? 1
- (ii) Show that $x = \frac{1}{2} \ln(y - 1)$ 1
- (iii) The volume V formed when the area between $y = 1 + e^{2x}$, the y -axis and the lines $y = 2$ and $y = 6$, is rotated about the y -axis is given by $V = \frac{\pi}{4} \int_2^6 [\ln(y - 1)]^2 \, dy$.
- Use Simpson's rule with 5 function values to estimate this volume.
Leave your answer rounded to 3 significant figures. 4

End of Section B.

Section C – Start a new booklet.**Question 5 (15 marks).****Marks**

- a)** Richard won the NSW Lotto on the 1st January 2000. The prize was one million dollars. He decided to deposit the entire amount into an account which paid interest at a rate of 8% per annum. The interest was calculated quarterly and compounded quarterly. Richard then made an annual withdrawal \$50000, starting on the 1st January 2001.

- (i) Write down an expression for the amount in Richard's account immediately after his first withdrawal. 2

- (ii) Show that the amount in Richard's account immediately after his third withdrawal is given by the expression: 3

$$10^6 \times 1.02^{12} - 50000(1 + 1.02^4 + 1.08^8)$$

- (iii) How much, to the nearest dollar, was left in Richard's account after his twentieth withdrawal. 1

- b)** Given that $f(x) = x(x-2)^2$

- (i) Show that $f'(x) = 3x^2 - 8x + 4$ 1

- (ii) Find 2 values of x for which $f'(x) = 0$ and give corresponding values of $f(x)$. 2

- (iii) Determine the nature of the turning points of the curve $y = f(x)$. 2

- (iv) Find where the curve $y = f(x)$ crosses the x -axis. 1

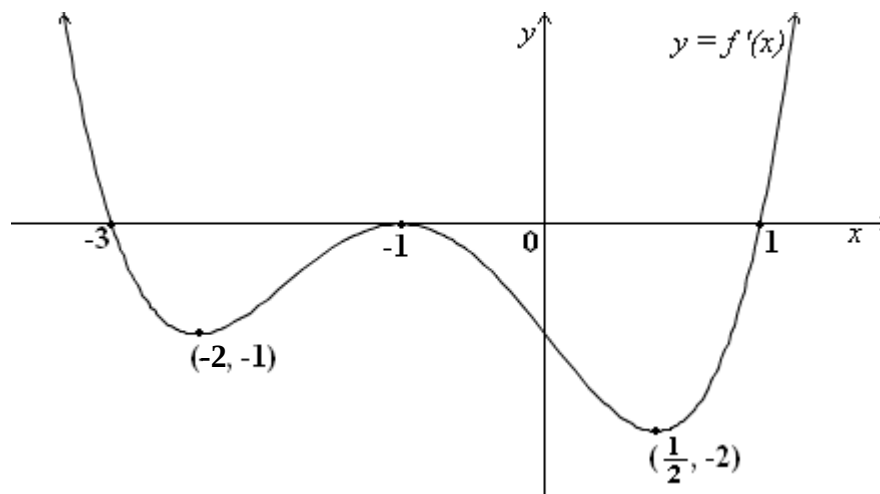
- (v) Sketch the curve $y = f(x)$. 2

- (vi) Use your sketch to solve the inequation $x(x-2)^2 \geq 0$ 1

Marks

Question 6 (14 marks).

a)



Consider the graph of $y = f'(x)$ shown above. Find the x co-ordinates of the minimum turning point and the maximum turning point of the graph $y = f(x)$.

3

b) Consider the two functions $f(x)$ and $g(x)$ where:

$$f(x) = \frac{e^x - e^{-x}}{2} \qquad g(x) = \frac{e^x + e^{-x}}{2}$$

- (i) Show that $f'(x) = g(x)$ and $g'(x) = f(x)$.
- (ii) Show that the graph $y = f(x)$ is increasing for all values of x and that there is a point of inflexion at $x = 0$.
- (iii) Show that the graph $y = g(x)$ has a minimum at $x = 0$.
- (iv) Sketch the graph of $y = g(x)$.
- (v) Let $y = f(x)$. Show that this equation can be written in the form:

$$e^{2x} - 2ye^x - 1 = 0$$

Hence deduce that $x = \ln\left(y + \sqrt{y^2 + 1}\right)$.

2

2

2

2

3

End of Section C.

End of Examination.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, x > 0$

$$1) a) i) \quad y = \ln 3x$$

$$y' = \frac{3}{3x}$$

$$y' = \frac{1}{x}$$

$$ii) \quad y = \tan \frac{x}{2}$$

$$y' = \frac{1}{2} \sec^2 \frac{x}{2}$$

$$iii) \quad y = \frac{1}{2} \sin 4x$$

$$y' = 2 \cos 4x$$

$$iv) \quad y = 4 + 5e^{-x}$$

$$y' = -5e^{-x}$$

$$b) i) \quad y = e^{0.2x}$$

$$y' = 0.2e^{0.2x}$$

$$y'' = 0.04e^{0.2x}$$

$$ii) \quad f(x) = 5 - 2 \ln x$$

$$f'(x) = -\frac{2}{x}$$

$$= -2x^{-1}$$

$$f''(x) = 2x^{-2}$$

$$= \frac{2}{x^2}$$

$$f''(2) = \frac{2}{(2)^2}$$

$$= \frac{1}{2}$$

$$iii) \quad y = x - 2x^3$$

$$y' = 1 - 6x^2$$

$$y'' = -12x$$

$$y'' < 0$$

$$-12x < 0$$

$$x > 0$$

$$iv) \quad y = 3x^3 - 3x^2 + 3x - 1$$

$$y' = 9x^2 - 6x + 3$$

$$y'' = 18x - 6$$

$$\text{let } y'' = 0$$

$$18x - 6 = 0$$

$$18x = 6$$

$$x = \frac{1}{3}$$

$$\text{when } x = \frac{1}{3},$$

$$y = 3\left(\frac{1}{3}\right)^3 - 3\left(\frac{1}{3}\right)^2 + 3\left(\frac{1}{3}\right) - 1$$

$$y = -\frac{2}{9}$$

the point of inflexion is at $\left(\frac{1}{3}, -\frac{2}{9}\right)$

$$c) \quad \int_{-1}^0 \sin \pi x \, dx$$

$$= \left[-\frac{1}{\pi} \cos \pi x \right]_{-1}^0$$

$$= -\frac{1}{\pi} \cos(0) - \left(-\frac{1}{\pi} \cos(-\pi) \right)$$

$$= -\frac{1}{\pi} - \frac{1}{\pi}$$

$$= -\frac{2}{\pi}$$

$$d) \quad y = x^2 - 4 \quad \text{--- (1)}$$

$$y = -3x \quad \text{--- (2)}$$

sub (1) into (2)

$$x^2 - 4 = -3x$$

$$x^2 + 3x - 4 = 0$$

$$(x+4)(x-1) = 0$$

$$x = -4, \quad x = 1$$

clearly $x = 1$

when $x = 1$

$$y = -3(1)$$

$$y = -3$$

$\therefore A$ is the point $(1, -3)$

$$\text{Area} = - \int_0^1 -3x \, dx - \int_1^2 (x^2 - 4) \, dx + \int_2^3 (x^2 - 4) \, dx$$

$$= \int_0^1 3x \, dx + \int_1^2 (4 - x^2) \, dx + \int_2^3 (x^2 - 4) \, dx$$

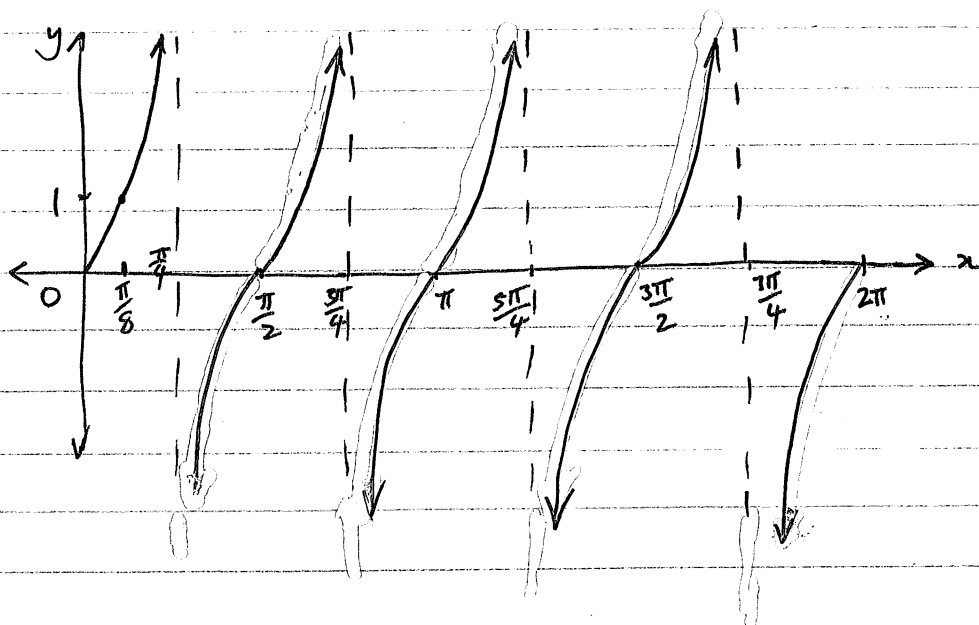
$$= \left[\frac{3x^2}{2} \right]_0^1 + \left[4x - \frac{x^3}{3} \right]_1^2 + \left[\frac{x^3}{3} - 4x \right]_2^3$$

$$= \left[\frac{3(1)^2}{2} - 0 \right] + \left[4(2) - \frac{(2)^3}{3} - \left(4(1) - \frac{(1)^3}{3} \right) \right] + \left[\frac{(3)^3}{3} - 4(3) - \left(\frac{(2)^3}{3} - 4(2) \right) \right]$$

$$= \frac{11}{2} \text{ units}^2$$

e) $y = \tan 2x$

period = $\frac{\pi}{2}$



$$2) a) \log_7 64 = 3 \log_7 x$$

$$\log_7 64 = \log_7 x^3$$

$$\therefore x^3 = 64$$

$$x = 4$$

$$b) \int_{\frac{1}{3}}^{\frac{1}{2}} \sec^2\left(\frac{\pi x}{2}\right) dx$$

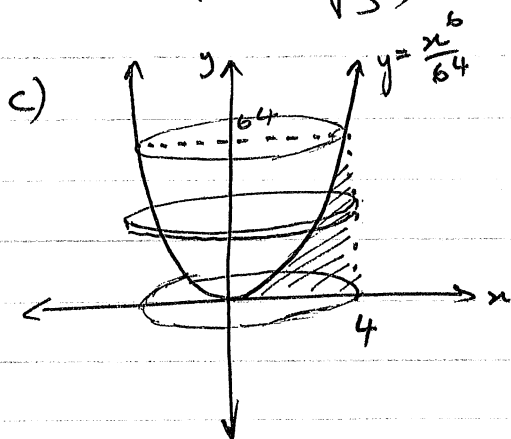
$$= \left[\frac{1}{\left(\frac{\pi}{2}\right)} \tan\left(\frac{\pi x}{2}\right) \right]_{\frac{1}{3}}^{\frac{1}{2}}$$

$$= \left[\frac{2}{\pi} \tan\left(\frac{\pi x}{2}\right) \right]_{\frac{1}{3}}^{\frac{1}{2}}$$

$$= \frac{2}{\pi} \tan\left(\frac{\pi}{4}\right) - \frac{2}{\pi} \tan\left(\frac{\pi}{6}\right)$$

$$= \frac{2}{\pi} (1) - \frac{2}{\pi} \left(\frac{1}{\sqrt{3}}\right)$$

$$= \frac{2}{\pi} \left(1 - \frac{1}{\sqrt{3}}\right)$$



when $x = 4$

$$y = \frac{(4)^6}{64}$$

$$y = 64$$

$$i) y = \frac{x^6}{64}$$

$$x^6 = 64y$$

$$x^2 = \sqrt[3]{64y}$$

$$x^2 = 4y^{\frac{1}{3}}$$

$$ii) V = \pi \int_a^b x^2 dy$$

$$V = \pi \int_0^{64} (4)^2 dy - \pi \int_0^{64} 4y^{\frac{1}{3}} dy$$

$$= 16\pi \int_0^{64} dy - 4\pi \int_0^{64} y^{\frac{1}{3}} dy$$

$$= 16\pi [y]_0^{64} - 4\pi \left[\frac{3}{4} y^{\frac{4}{3}} \right]_0^{64}$$

$$= 16\pi (64 - 0) - 3\pi ((64)^{\frac{4}{3}} - 0)$$

$$= 256\pi \text{ units}^3$$

$$d) 240^\circ \times \frac{\pi}{180} = \frac{4\pi}{3}$$

$$e) i) AY:YC = AX:XB$$

$$= 3:4$$

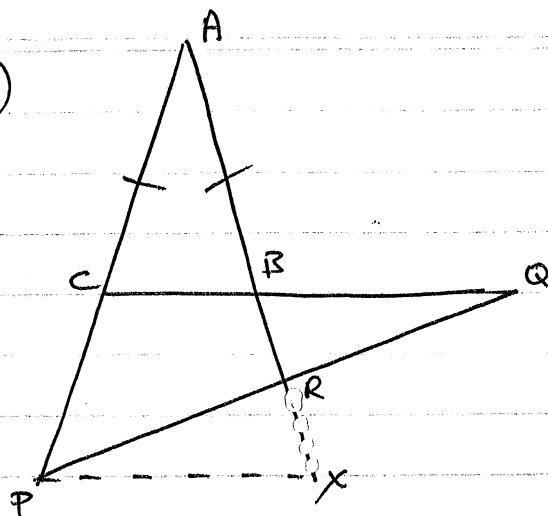
$$ii) AY:AC = AX:AB$$

$$= 3:7$$

$$iii) YC:AC = XB:AB$$

$$= 4:7$$

f)



X lies on AR produced
such that $PX \parallel CQ$.

In Δ 's BRQ & PRX

$$\angle BRQ = \angle PRX \text{ (vert. opp. } \angle\text{'s)}$$

$$\angle BQR = \angle RPX \text{ (alt } \angle\text{'s } BQ \parallel PX)$$

$\therefore BRQ \parallel \Delta PRX$ (equiangular)

$$\therefore \frac{BR}{RQ} = \frac{RX}{PR} \text{ (corr. sides in same ratio)}$$

$$BR \cdot PR = RX \cdot RQ$$

$$AB = AC \text{ (given)}$$

$$\text{clearly } AX = AP$$

$$\therefore PC = BX$$

$$= BR + RX$$

$$\therefore RX = PC - BR$$

$$BR \cdot PR = (PC - BR) RQ$$

$$BR \cdot PR = PC \cdot RQ - BR \cdot RQ$$

$$BR \cdot PR + BR \cdot RQ = PC \cdot RQ$$

$$BR (PR + RQ) = PC \cdot RQ$$

$$BR \cdot PQ = PC \cdot RQ$$

Q3 (a) (i).

x	0	$\frac{1}{2}$	1	$\frac{1}{4}$	2
y	1	$\frac{2}{3}$	$\frac{1}{2}$	$\frac{2}{5}$	$\frac{1}{3}$

(1)

$$(ii) \quad \frac{\frac{1}{2}}{2} \left[1 + \frac{1}{3} + 2 \left(\frac{2}{3} + \frac{1}{2} + \frac{2}{5} \right) \right]$$

$$= \frac{1}{4} \left[\frac{4}{3} + 2 \left(\frac{20 + 15 + 12}{30} \right) \right]$$

$$= \frac{1}{4} \left[\frac{4}{3} + \frac{47}{15} \right]$$

$$= \frac{1}{4} \frac{20 + 47}{15}$$

$$= \frac{67}{60}$$

$$= 1.12$$

(b) Smallest e^{-n}
Largest e^n .

$$(c) (i) \quad A = xy + \frac{1}{2} x^2 \sin 60^\circ$$

$$= xy + \frac{1}{2} x^2 \cdot \frac{\sqrt{3}}{2}$$

$$= xy + \frac{\sqrt{3} x^2}{4} = 33$$

$$\therefore xy = 33 - \frac{\sqrt{3} x^2}{4}$$

$$y = 33x^{-1} - \frac{\sqrt{3}}{4} x.$$

$$(ii) \quad P = 3x + 2y$$

$$= 3x + 66x^{-1} - \frac{\sqrt{3}}{2} x.$$

$$P' = 3 - \frac{\sqrt{3}}{2} - 66x^{-2}$$

$$P'' = 132x^{-3}$$

Let $P' = 0$ for max/min.

$$3 - \frac{\sqrt{3}}{2} = \frac{66}{x^2}$$

$$6 - \sqrt{3} = \frac{132}{x^2}$$

$$\frac{x^2}{132} = \frac{1}{6 - \sqrt{3}}$$

$$= \frac{6 + \sqrt{3}}{33}$$

$$\therefore x^2 = \frac{132(6 + \sqrt{3})}{33}$$

$$= 4(6 + \sqrt{3})$$

$$x = \pm 2\sqrt{6 + \sqrt{3}}$$

✓✓

Clearly $x > 0$ & a min occurs
as $P'' > 0$

✓

$$\text{at } x = 2\sqrt{6 + \sqrt{3}}$$

Q1. (i) $P = r + r + r\theta$

$$= 2r + r\theta$$

$$= r(2 + \theta)$$

✓✓

(ii) If $P = 36$.

now $A = \frac{1}{2} r^2 \theta$

$$r(2 + \theta) = 36$$

$$= \frac{1}{2} \left(\frac{36}{2 + \theta} \right)^2 \cdot \theta$$

$$r = \frac{36}{2 + \theta}$$

$$A = \frac{648\theta}{(2 + \theta)^2}$$

✓✓

$$(iii) \quad A = \frac{648\theta}{(2+\theta)^3}$$

$$A' = \frac{(2+\theta)^2 \cdot 648 - 648\theta \cdot 2(2+\theta)}{(2+\theta)^4}$$

$$= \frac{648(2+\theta)[2+\theta-2\theta]}{(2+\theta)^4}$$

$$= \frac{648(2-\theta)}{(2+\theta)^3}$$

✓✓

$$\text{Let } A' = 0$$

$$\frac{648(2-\theta)}{(2+\theta)^3} = 0$$

$$\theta = 2.$$

$$\therefore A = \frac{648 \times 2}{4^2} = \frac{648}{8} = 81. \quad \checkmark$$

θ	1	2	3
A'	$\frac{648}{3^3}$ \$>0\$	0	$-\frac{648}{5^3}$ \$<0\$
	/	-	\

$\therefore$ MAX.

✓

NB The "2nd derivative test" is probably not the best option.

Also, $\begin{matrix} / & - & \backslash \\ + & 0 & - \end{matrix}$ is

not sufficient (we were told it is a max) & you need to show it !!

Q4.

a

$$\frac{d}{dx}(x^r \ln x) = 2x \ln x + x^r \times \frac{1}{x} \\ = 2x \ln x + x. \quad \checkmark$$

$$\therefore \int_{\sqrt{e}}^e (2x \ln x + x) dx = \left[x^2 \ln x \right]_{\sqrt{e}}^e \\ = e^r \ln e - (\sqrt{e})^r \ln \sqrt{e} \\ = e^r - e. \times \frac{1}{2} \\ = e^r - \frac{e}{2}.$$

$$\therefore \int_{\sqrt{e}}^e 2x \ln x dx + \int_{\sqrt{e}}^e x dx = e^r - \frac{e}{2}$$

$$2 \int_{\sqrt{e}}^e x \ln x dx + \left[\frac{x^2}{2} \right]_{\sqrt{e}}^e = e^r - \frac{e}{2}$$

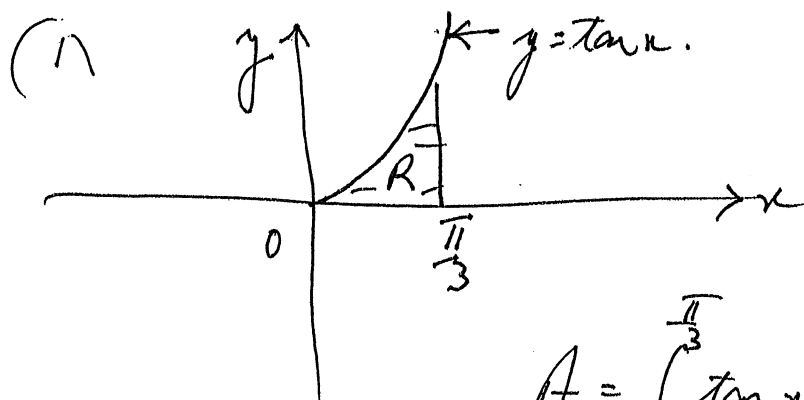
$$2 \int_{\sqrt{e}}^e x \ln x dx + \frac{e^r}{2} - \frac{(\sqrt{e})^r}{2} = e^r - \frac{e}{2}$$

$$2 \int_{\sqrt{e}}^e x \ln x dx + \frac{e^r}{2} - \frac{e}{2} = e^r - \frac{e}{2}$$

$$2 \int_{\sqrt{e}}^e x \ln x dx = \frac{e^r}{2} \quad \checkmark \checkmark \checkmark$$

$$\therefore \int_{\sqrt{e}}^e x \ln x dx = \frac{e^r}{4}$$

Q4 b
(cont)



$$\begin{aligned}
 A &= \int_0^{\frac{\pi}{3}} \tan x \, dx \\
 &= \int_0^{\frac{\pi}{3}} \frac{\sin x}{\cos x} \, dx \\
 &= -\left[\ln(\cos x) \right]_0^{\frac{\pi}{3}} \\
 &= -\left[\ln \frac{1}{2} - \ln 1 \right] \\
 &= \ln 2 \quad \checkmark
 \end{aligned}$$

(ii)

$$\begin{aligned}
 V &= \pi \int_0^{\frac{\pi}{3}} \tan^2 x \, dx \\
 &= \pi \int_0^{\frac{\pi}{3}} (\sec^2 x - 1) \, dx \\
 &= \pi \left[\tan x - x \right]_0^{\frac{\pi}{3}} \\
 &= \pi \left[\tan \frac{\pi}{3} - \frac{\pi}{3} - (0 - 0) \right] \\
 &= \pi \cdot \left(\sqrt{3} - \frac{\pi}{3} \right) \quad \checkmark \checkmark
 \end{aligned}$$

Q4 (CONTD)

c (i) $x \in \mathbb{R}$. (all Reals etc)

(ii) $y = 1 + e^{2x}$

$$y - 1 = e^{2x}$$

$$\therefore 2x = \ln(y - 1)$$

$$\boxed{x = \frac{1}{2} \ln(y - 1)}$$

(iii) Using Simpson's Rule. to evaluate the integral

$$\frac{\pi}{4} \int_2^6 [\ln(y-1)]^2 dy. \quad (\text{let } f(y) = [\ln(y-1)]^2)$$

y	2	3	4	5	6
f(y)	$(\ln 1)^2$	$(\ln 2)^2$	$(\ln 3)^2$	$(\ln 4)^2$	$(\ln 5)^2$
	= 0	= 0.4804	= 1.2069	= 1.9218	= 2.5902

$$\therefore V = \frac{\pi}{4} \times \frac{1}{3} \left[0 + 2.5902 + 4(0.4804 + 1.9218) + 2(1.2069) \right]$$

$$\hat{=} 3.83 \text{ m}^3 \quad (\text{to 3. SIG FIGS})$$

Task 3 2007 2 unit Solutions.

5 (a) January 1st 2000 \$1,000,000

$$r = 8\% \text{ p.a.} \Rightarrow \frac{8}{4} = 2\% \text{ per quarter}$$

interest calculated quarterly, paid quarterly.

Annual withdrawal: \$50,000 starting Jan 1 2001.

$$(i) \text{ Jan 1 2001 } A_1 = 1,000,000 \left(1 + \frac{2}{100}\right)^4 - 50,000$$

$$= 10^6 \times 1.02^4 - 50,000 \quad (2)$$

$$(ii) \text{ Jan 1 2002 } A_2 = (10^6 \times 1.02^4 - 50,000) \left(1 + \frac{2}{100}\right)^4 - 50,000$$

$$= (10^6 \times 1.02^4 - 50,000) (1.02)^4 - 50,000$$

$$= 10^6 \times 1.02^8 - 50,000 - 50,000 \times 1.02^4$$

$$= 10^6 \times 1.02^8 - 50,000(1 + 1.02^4)$$

$$\text{Jan 1 2003 } A_3 = 10^6 \times 1.02^{12} - 50,000(1 + 1.02^4 + 1.02^8)$$

(3)

$$(iii) 20^{\text{th}} \text{ withdrawal}$$

$$A_{20} = 10^6 \times 1.02^{80} - 50,000(1 + 1.02^4 + 1.02^8 + \dots + 1.02^{76})$$

$$= 10^6 \times 1.02^{80} - 50,000 \left(\frac{1 - 1.02^4 \times 1.02^{76}}{1 - 1.02^4} \right)$$

$$= 1,000,000 \times 1.02^{80} - 50,000 \left(\frac{1 - 1.02^{80}}{1 - 1.02^4} \right)$$

$$= 4875439.156 - 2350683.978$$

$$= \$2524755.18$$

$\therefore$ \$2524755 was left.

(1)

using
 $\frac{a - rl}{1 - r^n}$



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5 (b) $f(x) = x(x-2)^2$

(i) $f(x) = x(x-2)(x-2)$
 $= x(x^2 - 4x + 4)$
 $= x^3 - 4x^2 + 4x$

$f'(x) = 3x^2 - 8x + 4$ (1)

(ii) When $3x^2 - 8x + 4 = 0$ $x = 12$
 $(3x-6)(3x-2) = 0$ $+ -8$

$3(x-2)(3x-2) = 0$

$x = 2$ $x = \frac{2}{3}$

$f(x) = 2(2-2)^2 = 0$

$f(x) = \frac{2}{3}(\frac{2}{3}-2)^2$

$= \frac{2}{3} \times -\frac{4}{3} \times -\frac{4}{3} = 1\frac{5}{27}$

$(2, 0)$ (1)

$(\frac{2}{3}, 1\frac{5}{27})$ (1)

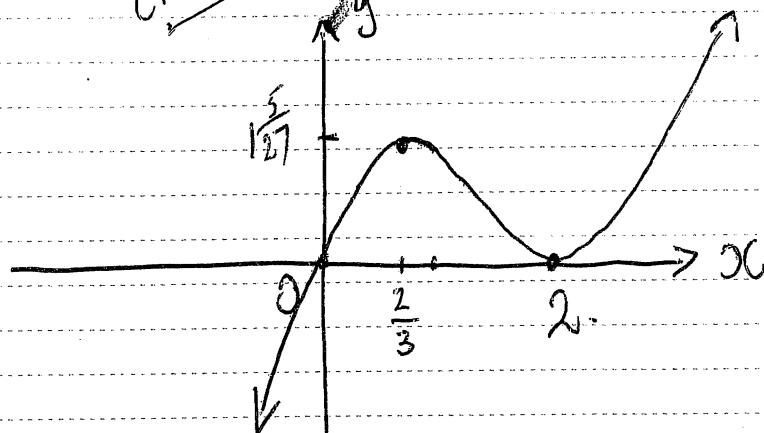
(iii) now $f'(x) = 3x^2 - 8x + 4$
 $f''(x) = 6x - 8$

At $(2, 0)$ $f''(x) = 12 - 8 = 4 > 0$ min t. pt (1)

At $(\frac{2}{3}, 1\frac{5}{27})$ $f''(x) = 6 \times \frac{2}{3} - 8 = 4 - 8 = -4 < 0$
 MAX t. pt (1)

(iv) $f(x) = x(x-2)^2$
 when $f(x) = 0$ $\xrightarrow{\text{crosses}} x = 0, x = 2$ (double root) (1)

(v)



check, $x=3$ $f(x) = 3(3-2)^2 = 3$.

$x=-1$, $f(x) = -1(-1-2)^2 = -1 \times 9 = -9$

(vi) $x(x-2)^2 \geq 0 \Rightarrow x \geq 0$ from sketch
 (1)



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Q6 (a) $f'(x)$ gives gradient of tangent lines at x values. eg $(-3, -1)$ means at $x = -3$ on the original curve, the slope of the tangent line at $x = -3$ is $m = -1$.

Minimum turning pt

$x - \epsilon$	$\backslash$	negative slope
x	0	
$x + \epsilon$	$/$	positive slope

$x = 1 - \epsilon$ slope negative

$x = 1$ slope is zero

$x = 1 + \epsilon$ slope positive

MAX. turning point

$x - \epsilon$	$/$	positive slope
x	0	
$x + \epsilon$	$\backslash$	negative slope

$x = -3 - \epsilon$ slope +ve

$x = -3$ slope 0

$x = -3 + \epsilon$ slope -ve

$1\frac{1}{2}$

$1\frac{1}{2}$

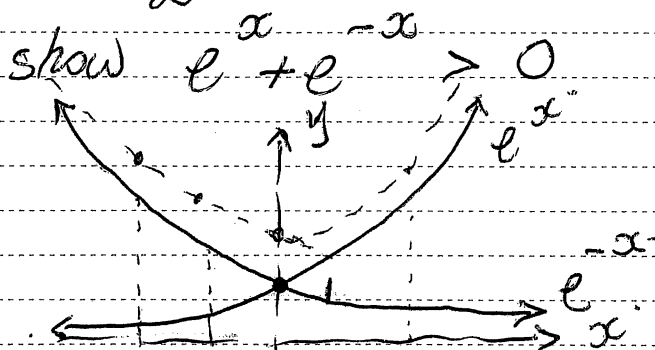
$$6(b) \quad f(x) = \frac{e^x - e^{-x}}{2} \quad g(x) = \frac{e^x + e^{-x}}{2}$$

$$(i) \quad f'(x) = \frac{e^x - e^{-x} \cdot (-1)}{2} = \frac{e^x + e^{-x}}{2} = g(x)$$

and $g'(x) = \frac{e^x + e^{-x} \cdot (-1)}{2} = \frac{e^x - e^{-x}}{2} = f(x). \quad (2)$

(ii) If $y = f(x)$ is an increasing graph then $f'(x) > 0 \quad \forall x$.

show $\frac{e^x + e^{-x}}{2} > 0 \quad \forall x$.



The addition of ordinates of $e^x, e^{-x} \quad \forall x$ will always be > 0 . (1)

$$f'(x) = \frac{e^x + e^{-x}}{2}$$

$$f''(x) = \frac{e^x + e^{-x} \cdot (-1)}{2} = \frac{e^x - e^{-x}}{2}$$

So at $x = 0 - \epsilon$ $f''(x) = \frac{e^{-\epsilon} - e^{\epsilon}}{2} < 0$
(eg -0.1)

$x = 0$ $f''(x) = \frac{1-1}{2} = 0$

$x = 0 + \epsilon$ $f''(x) = \frac{e^{\epsilon} - e^{-\epsilon}}{2}$
(eg 0.1)

sign change > 0
Inflection at $x = 0$. (1)

$$(iii) \quad g(x) = \frac{e^x + e^{-x}}{2}$$

$$g'(x) = \frac{e^x - e^{-x}}{2}$$

$$\text{When } \frac{e^x - e^{-x}}{2} = 0$$

$$\frac{e^x}{1} - \frac{1}{e^x} = 0$$

$$\frac{e^x}{1} = \frac{1}{e^x}$$

$$(e^x)^2 = 1$$

$$e^{2x} = e^0$$

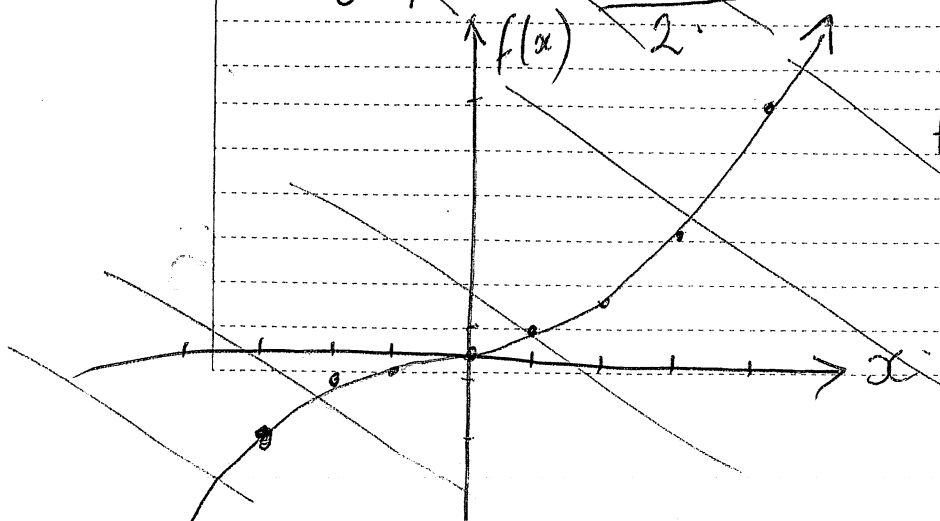
$$2x = 0 \Rightarrow x = 0$$

$$g''(x) = \frac{e^x + e^{-x}}{2}$$

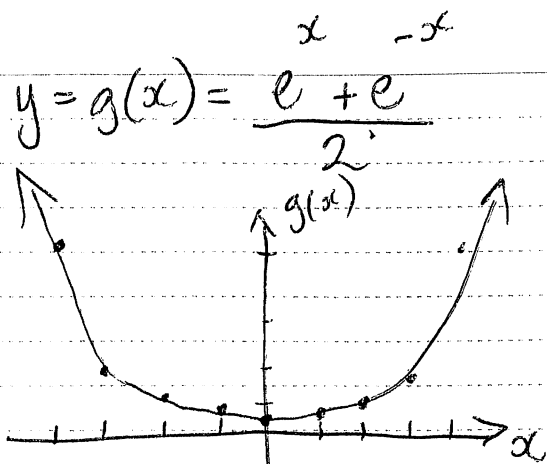
$$\text{at } x=0, \quad g''(x) = \frac{e^0 + e^{-0}}{2} = \frac{1+1}{2} > 0 \text{ min} \quad (2)$$

$$(iv) \quad y = f(x) = \frac{e^x - e^{-x}}{2}$$

x	-4	-3	-2	-1	0	1	2	3	4
$f(x)$	-27.3	-10	-3.6	-1.2	0	1.2	3.6	10	27.3



(1)



x	-4	-3	-2	-1	0	1	2	3	4
$g(x)$	27.3	10	3.8	1.5	1	1.5	3.8	10	27.3

(2)

(b) $y = \frac{e^x - e^{-x}}{2}$

$$2y = e^x - \frac{1}{e^x}$$

$$2ye^x = e^{2x} - 1$$

$$e^{2x} - 2ye^x - 1 = 0$$

now for $e^{2x} - 2ye^x - 1 = 0$
let $w = e^x$

so $w^2 - 2wy - 1 = 0$
 $a=1, b=-2y, c=-1$

$$w = \frac{2y \pm \sqrt{4y^2 - 4 \times 1 \times -1}}{2}$$

$$= \frac{2y \pm \sqrt{4y^2 + 4}}{2}$$

$$= \frac{2y \pm 2\sqrt{y^2 + 1}}{2}$$

$$w = y \pm \sqrt{y^2 + 1}$$

now $e^x = y \pm \sqrt{y^2 + 1}$

$$\ln e^x = \ln(y \pm \sqrt{y^2 + 1})$$

$$x = \ln(y \pm \sqrt{y^2 + 1})$$

y can be any value so for $\ln$ to exist
 $y + \sqrt{y^2 + 1} \geq 0$ is chosen.

So $x = \ln(y + \sqrt{y^2 + 1})$

(3)